



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · 4 product categories

Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating column

\$59.76

Avg Purchase

Demographics · purchase details · shopping behavior · subscription status

Data Preparation in Python

Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case

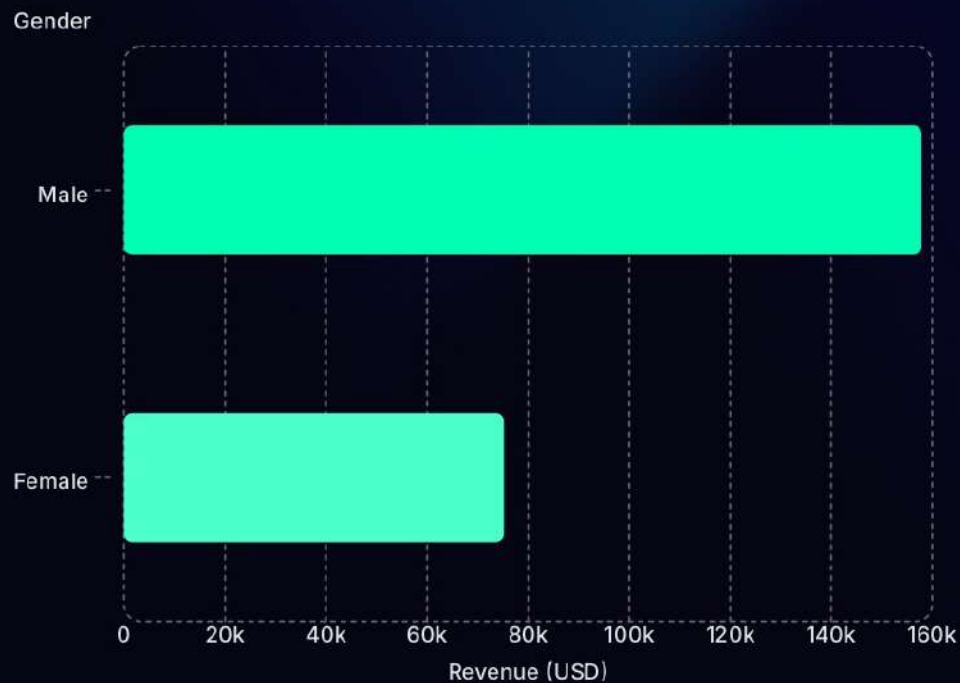
Feature Engineering

Created **age_group** and **purchase_frequency_days**; dropped redundant **promo_code_used**

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for SQL analysis

Revenue by Gender



Key Insight

Male customers generated **2.1x more revenue** than female customers — a critical segment for targeted campaigns.

Subscribers vs. Non-Subscribers



Subscription Snapshot

73% non-subscribers vs **27%** subscribers

Average spend nearly identical: \$59.87 vs \$59.49

- ❏ Subscribers are fewer but represent a high-value retention opportunity.

Top Products & Discount Dependency

Top 5 by Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rate

Item	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

CUSTOMER SEGMENTATION



Customer Segmentation

Loyal

3,116 customers

Returning

701 customers

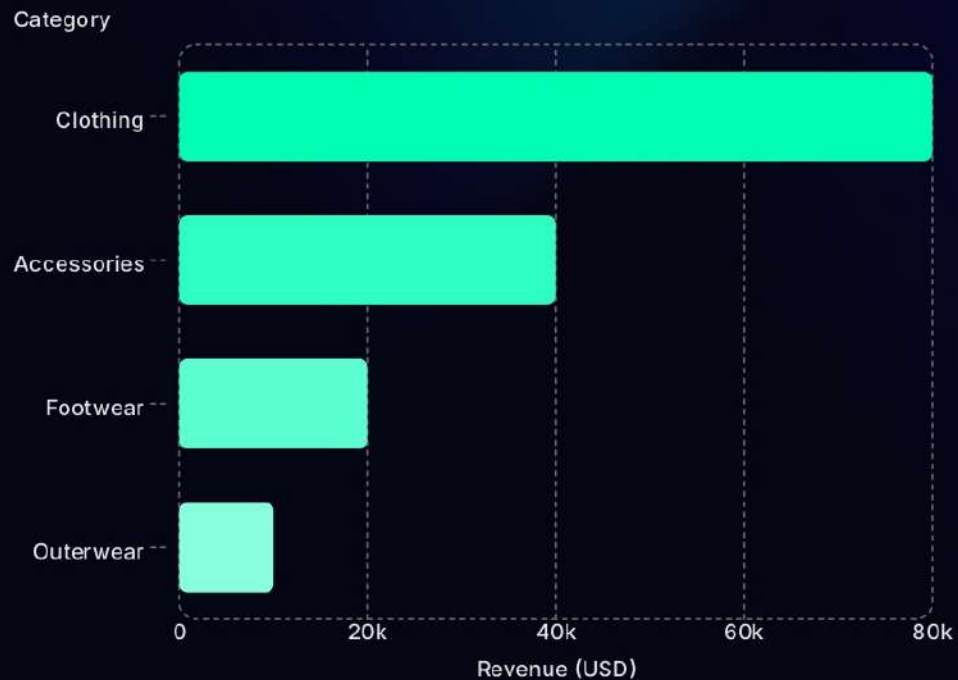
New

83 customers

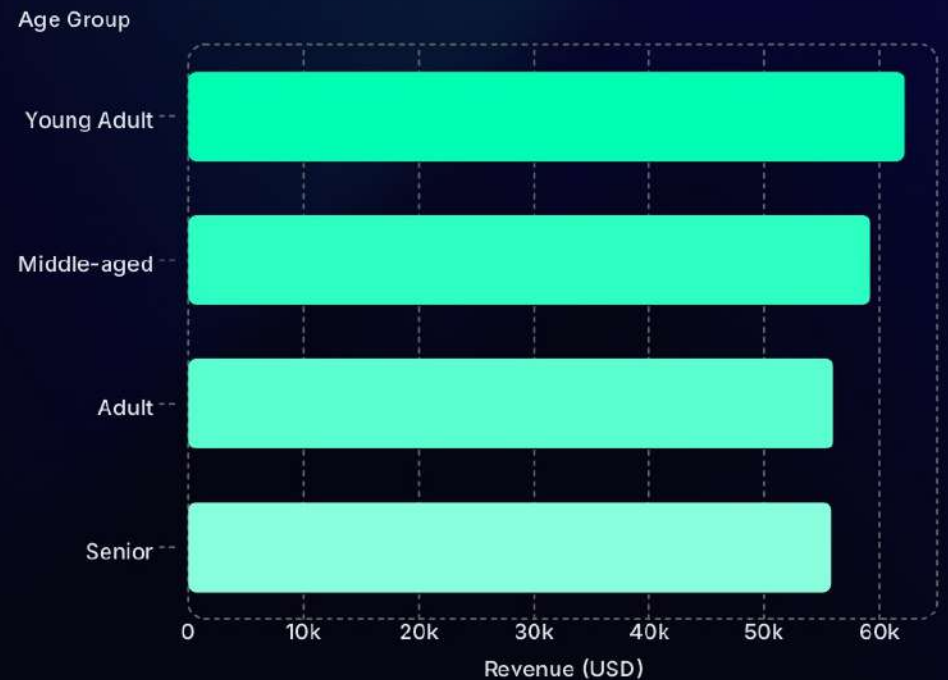
839 customers used discounts yet spent above average — high-value discount users worth nurturing.

Revenue by Category & Age Group

Revenue by Category



Revenue by Age Group



- Filters: subscription, gender, category, shipping type
- 3.9K customers · \$59.76 avg purchase · 3.75 avg rating
- Revenue & sales by category and age group
- 27% subscribers vs 73% non-subscribers

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Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users



Product Positioning

Highlight top-rated & best-selling products in campaigns